Ling Wang

School of Mathematical Science, Peking University
No. 5 Yiheyuan Road, Haidian District, Beijing, P.R.China 100871
Office: Room 203, Building 19, Peking University
E-mail: lingwang@stu.pku.edu.cn
Personal website: lwmath.github.io

Research interests

Geometric Analysis & Nonlinear PDEs, especialy Monge-Ampère type equations.

Education

2020.9 – Present **Peking University** – Beijing, China
PhD in pure mathematics
Mentors: Professor Bin Zhou.

2020.2 – 2020.6 **Beijing International Center for Mathematical Research** – Beijing, China
Visiting student, pure mathematics.

2016.9 – 2020.6 **Beijing Normal University** – Beijing, China
BA in mathematics and applied mathematics
Mentors: Professor Haigang Li.

Publications

[6] **L. Wang**. Interior Hölder regularity of the linearized Monge-Ampère equation. To appear in *Calc. Var. Partial Differential Equations*.

[5] **L. Wang**, B. Zhou. $C^{1,\alpha}$ regularity of variational problems with a convexity constraint. Preprint.

[4] **L. Wang**, B. Zhou. Liouville theorems for a class of degenerate or singular Monge-Ampère equations. *J. Geom. Anal.*, **34** (2024), no. 11, Paper No. 352.

[3] Y. H. Kim, N. Q. Le, **L. Wang**, B. Zhou. Singular Abreu equations and linearized Monge-Ampère equations with drifts. To appear in *J. Eur. Math. Soc. (JEMS)*.

[2] **L. Wang**, B. Zhou. Interior estimates for Monge-Ampère type fourth order equations. *Rev. Mat. Iberoam.*, **39** (2023), no. 5, 1895–1923.

[1] H. G. Li, J. -N. Wang, **L. Wang**. Refined stability estimates in electrical impedance tomography with multi-layer structure. *Inverse Probl. Imaging*, **16** (2022), no. 1, 229–249.

Honors and scholarships

2024 China National Scholarship

Presidential Scholarship (Peking University)

Outstanding Research Award (Peking University)

2023 Presidential Scholarship (Peking University)

Merit Student (Peking University)

2022 Exceptional Award for Academic Innovation (Peking University)

Talks

July 21-27, 2024 Workshop on Geometric Analysis 2024, School of Mathematical Sciences of Inner Mongolia University, Hohhot, China

March 24, 2024 Ph.D. Mathematics Forum, School of Mathematics and Statistics of Wuhan University, Wuhan, China

March 30, 2022 Geometric PDE seminar, Academy of Mathematics and Systems Science of the Chinese Academy of Sciences, Beijing, China

Teaching experience

Spring 2024 **Teaching assistant, Advanced math B (PKU)**

Fall 2023 **Teaching assistant, Mathematical analysis III (PKU)**

Spring 2023 **Teaching assistant, Mathematical analysis II (PKU)**

Fall 2022 **Teaching assistant, Mathematical analysis I (PKU)**

Spring 2022 **Teaching assistant, Arithmetic of elliptic curves (PKU)**
Teaching assistant, Diophantine approximation (PKU)

Fall 2021 **Teaching assistant, Mathematical analysis I (PKU)**

Spring 2021 **Teaching assistant, Functional analysis (PKU)**

Fall 2020 **Teaching assistant, Advanced math C (PKU)**